Horizontal gene transfer emerges at scale: critical background escape and the control of antibiotic-resistance plasmids

Abstract

Horizontal gene transfer (HGT) is usually explained by its immediate capacity to acquire useful genes, yet conjugative transfer can be costly and its benefit depends strongly on donor, recipient and environment. We develop a scale-dependent theory in which plasmid HGT is treated as directional background escape: vertical inheritance traps a plasmid in the evolutionary future of its current host, whereas conjugation releases it into another chromosomal background. Extending recent theories of sex, we show that the escape component of HGT retains a positive critical singularity when opposing plasmid and host fitness effects nearly cancel. If transfer accessibility remains finite near this cancellation surface, the distribution of episode-integrated HGT advantage has the universal tail $\mathbb{P}(V_{\mathrm{HGT}} > v) \sim v^{-1}$. Host-range restriction can weaken or eliminate the singularity in a predictable way. A binary spin representation gives the same result as critical scaling near maximum configurational entropy. We then translate the theory into a surveillance framework for antibiotic-resistance plasmids, combining host-resolved plasmid tracking, conjugation opportunity and finite-time escape scores. The framework predicts that control should target rare high-value transfer opportunities rather than mean conjugation alone.

Keywords: horizontal gene transfer; plasmid; antibiotic resistance; evolution of sex; conjugation; criticality; heavy tails; genomic surveillance

1 Background

Horizontal gene transfer (HGT) is one of the defining features of bacterial evolution. Through conjugation, transformation and transduction, genetic material can move between lineages that do not share a recent parent–offspring relationship [1–3]. For antibiotic resistance, the consequences are immediate: a resistance determinant can spread not only by clonal expansion of the bacterial lineage in which it arose, but also by movement among lineages and species. Conjugative plasmids are especially important because they are simultaneously genetic elements, vehicles of adaptive genes and transmissible evolutionary lineages in their own right [4–6].

The evolutionary advantage of HGT is nevertheless less trivial than its epidemiological importance might suggest. Conjugation requires cellular machinery, expression of transfer genes and contact with suitable recipients; plasmids can impose growth costs; and the success of a transfer depends strongly on the recipient background and environment [5, 7–9]. Thus, "HGT is advantageous because it spreads useful genes" is incomplete in the same way that "sex is advantageous because it creates variation" is incomplete. Both statements identify a possible consequence of genetic exchange without specifying when selection should favour the machinery that produces it.

A useful starting point comes from the theory of sex. In asexual reproduction, a genetic element is inherited together with its current background. If one component of that background is beneficial and another is deleterious, physical linkage forces them to share a common selective future. Recombination can break this association. Classical treatments express this benefit in terms of linkage disequilibrium, Hill–Robertson interference or related forms of selection interference [10–14]. Recent work takes a complementary, path-ensemble view: the relevant quantity is the selection accumulated by a linkage-breaking modifier over a complete selective episode, rather than the instantaneous advantage in a particular microscopic configuration [15]. Under weak assumptions, this integrated "sex potential" is non-negative in ensemble expectation and exhibits a heavy positive tail generated by near-cancellation of additive fitness effects. A second formulation using binary fitness effects shows that the same singularity occurs at maximum configurational entropy, where selective degeneracy, maximally negative aggregate covariance and critical slowing down coincide [16].

Here we ask what survives when sex is replaced by plasmid-mediated HGT. The mapping is close but not exact. Conjugation is directional rather than reciprocal; the donor usually persists after transfer; transfer is constrained by host range, contact and establishment; and horizontal transmission provides a direct reproductive benefit to the plasmid even before any advantage of escaping its present background is counted. These differences require modifications to the theory. They also make the theory experimentally useful, because the same quantities that modify the singularity—recipient accessibility, transfer opportunity, establishment and duration—are measurable targets for genomic surveillance and intervention.

Our central claim is therefore narrower than "HGT is sex". The *background-escape component* of HGT lies in the same critical universality class as recombination when transfer accessibility does not vanish at selective degeneracy. HGT-specific ecology can alter the amplitude, steepen the tail, or remove the singularity, but it does so through a small number of identifiable factors. This yields both a theory of evolutionary advantage and a practical framework for identifying the rare plasmid–host transitions most likely to drive antibiotic-resistance spread.

2 HGT as directional background escape

2.1 Two fitness components and one transferable association

Consider two competing plasmid–host associations. Let X_i denote the additive contribution associated with plasmid state i and Y_i the additive contribution of its chromosomal background,

Figure 1 placeholder.

A conceptual diagram contrasting reciprocal recombination with directional plasmid conjugation. Show two host backgrounds carrying alternative plasmid states; reciprocal sex creates both recombinant associations, whereas conjugation creates one new plasmid–host association while retaining the donor. Below, show the cancellation surface $\Delta_X + \Delta_Y = 0$ and the HGT accessibility weight ω on a directed host-transfer graph.

Figure 1: HGT as directional background escape. Vertical inheritance keeps a plasmid linked to its current chromosomal background. Conjugation creates a new plasmid–host association without requiring reciprocal exchange. The sex-like component of the advantage is the value of escaping the current background; the direct production of an additional plasmid-bearing cell is a separate contribution. Host range, contact and establishment determine which background-escape events are accessible.

with total Malthusian fitness

$$W_i = X_i + Y_i, \qquad i \in \{1, 2\}. \tag{1}$$

The plasmid state may represent two plasmid alleles, plasmid presence versus absence, or two mobile genetic backgrounds carrying alternative variants of a focal determinant. Define

$$\Delta_X = X_2 - X_1, \qquad \Delta_Y = Y_2 - Y_1, \qquad U = \Delta_X + \Delta_Y. \tag{2}$$

Thus $U = W_2 - W_1$ is the total selective difference between the two associations.

When Δ_X and Δ_Y have opposite signs, the plasmid component favoured by selection is initially coupled to the chromosomal background disfavoured by selection, or vice versa. Genetic exchange can then create a better matched association. In the reciprocal two-locus problem, the instantaneous advantage of association breaking is proportional to $-\operatorname{cov}(X, Y)$. In a two-genotype population, this covariance is proportional to $\Delta_X \Delta_Y p(1-p)$, where p is one genotype's frequency. Hence negative covariance favours breakup of the association [11, 12, 15].

Conjugation differs because it usually creates only one of the reciprocal associations. A donor carrying plasmid state 2 can transfer it to a recipient with background 1, producing (X_2, Y_1), while the donor (X_2, Y_2) remains. We therefore write the sex-like escape component of HGT potential as

$$V_{\text{esc}}(T) = -2q\,\omega(U, \Delta_X; \mathcal{E})\,\Delta_X \Delta_Y\,K_T(|U|), \tag{3}$$

where q is the initial frequency factor inherited from the two-lineage dynamics, $\omega \geq 0$ is an *accessibility weight*, $\mathcal{E}$ denotes ecological context, and K_T is a positive decreasing dynamical kernel. The weight ω absorbs the principal differences between reciprocal sex and HGT: directionality, donor–recipient contact, host range, transfer compatibility and successful establishment. Normalising reciprocal exchange to $\omega = 1$ is convenient, but no universal numerical factor is implied for conjugation.

For equal initial frequencies and deterministic selection, a useful finite-horizon kernel is

$$K_T(s) = \frac{\tanh(sT/2)}{s}, \qquad K_T(0) = \frac{T}{2}, \tag{4}$$

which tends to $1/s$ as $T \to \infty$. More generally, the results below require only that $K(s)$ be positive and decreasing and that, for a complete selective episode,

$$K(s) \sim \frac{c}{s} \qquad (s \downarrow 0), \tag{5}$$

with $c > 0$.

Equation (3) deliberately excludes the direct reproductive gain produced when conjugation makes an additional plasmid-bearing cell. That direct horizontal-transmission term is real and important, but it is not the analogue of the advantage of sex. Separating the two prevents a trivial transmission advantage from obscuring the more general evolutionary value of background escape.

2.2 The cancellation surface and the positive singularity

Set $Z = \Delta_X$. Because $\Delta_Y = U - Z$,

$$-\Delta_X \Delta_Y = Z^2 - UZ. \tag{6}$$

Near the cancellation surface $U = 0$, the numerator therefore tends to $Z^2 \geq 0$. If the HGT accessibility weight is finite and nonzero there,

$$\omega(U, Z; \mathcal{E}) \longrightarrow \omega_0(Z; \mathcal{E}) > 0, \tag{7}$$

then equations (3)–(5) give

$$V_{\text{esc}} \sim 2qc\,\omega_0(Z; \mathcal{E})\,\frac{Z^2}{|U|} \qquad (U \to 0). \tag{8}$$

The divergence is necessarily positive. Selective degeneracy of total association fitness therefore does not imply neutrality with respect to HGT: the two associations can have nearly identical total fitness precisely because large, opposing plasmid and host contributions cancel.

This is the same mechanism that produces the positive singularity for recombination, but the HGT interpretation is especially direct. When two plasmid–host associations have similar total fitness, neither rapidly excludes the other. The opportunity for transfer between their components persists for an anomalously long time. If the components themselves differ substantially, a successful plasmid transfer can reveal fitness that was hidden by their original packaging. Near degeneracy, favourable sign and long persistence arise from the same condition.

2.3 A universal tail, and how host range can change it

Let (U, Z) have a joint density continuous near $U = 0$. If

$$0 < M_{\mathrm{HGT}} := \int \omega_0(z; \mathcal{E}) z^2 f_{U,Z}(0, z)\, dz < \infty, \tag{9}$$

then the region producing $V_{\mathrm{esc}} > v$ has width proportional to $1/v$ around $U = 0$. Consequently,

$$\boxed{\mathbb{P}(V_{\mathrm{esc}} > v) \sim \frac{C_{\mathrm{HGT}}}{v}} \qquad (v \to \infty), \tag{10}$$

for a positive constant C_{HGT} determined by initial frequencies, the kernel and transfer accessibility. Directionality changes the prefactor through ω but not the exponent.

This conclusion has an important HGT-specific qualification. Transfer accessibility need not remain finite near the cancellation surface. Suppose ecological or molecular constraints make

$$\omega(U, Z; \mathcal{E}) \sim a(Z; \mathcal{E})|U|^{\gamma}, \qquad \gamma \geq 0. \tag{11}$$

For $0 \leq \gamma < 1$,

$$V_{\mathrm{esc}} \sim \frac{A(Z; \mathcal{E})}{|U|^{1-\gamma}}, \tag{12}$$

so

$$\boxed{\mathbb{P}(V_{\mathrm{esc}} > v) \sim C_{\gamma} v^{-1/(1-\gamma)}.} \tag{13}$$

At $\gamma = 1$ the singularity is regularised, and for $\gamma > 1$ the critical contribution vanishes at degeneracy. Thus the canonical $1/v$ tail is not a claim that plasmid biology is irrelevant. Rather, it is a null universality class: departures from slope -1 diagnose a systematic coupling between the fitness cancellation surface and the accessibility of transfer.

This distinction also clarifies which result from the sex theory is most robust. Under coordinate-reflection symmetry and a non-increasing kernel, reciprocal association breaking has non-negative ensemble mean potential [15]. HGT ecology can break the required reflection symmetry: a host background that is easy to enter need not be equally easy to leave, and recipient availability can correlate with the signs of fitness effects. Therefore a global theorem that $\mathbb{E}[V_{\mathrm{esc}}] \geq 0$ does not hold without additional symmetry assumptions on ω. The *positive critical tail*, however, survives any smooth positive accessibility at $U = 0$. HGT can therefore be locally or even typically costly while still experiencing rare episodes of very strong positive selection.

3 The binary spin limit

The same result can be expressed without continuous fitness effects. Suppose two leading plasmid–host associations differ across L additive components and write the signed component differences as

$$\sigma_\ell \in \{-1, +1\}, \qquad S = \sum_{\ell=1}^{L} \sigma_\ell. \tag{14}$$

Here S is simultaneously the total selective difference between the two associations and the magnetisation of the binary configuration [16]. The number of configurations with a given S is

$$\Omega_L(S) = \binom{L}{(L+S)/2}, \tag{15}$$

and the configurational entropy

$$\mathcal{H}(S) = \log \Omega_L(S) \tag{16}$$

is maximal at $S = 0$. For large L near the entropy maximum,

$$\epsilon := \mathcal{H}_{\max} - \mathcal{H} \simeq \frac{S^2}{2L}. \tag{17}$$

For reciprocal association breaking and equal initial frequencies, the integrated potential of a complete episode is

$$V_L = \frac{L - S^2}{2|S|} \sim \frac{L}{2|S|} \qquad (S \to 0). \tag{18}$$

The divergence is produced by two coincident facts: aggregate covariance is maximally negative at $S = 0$, and the relaxation time of competition between the two associations diverges as $|S|^{-1}$ [16]. HGT multiplies this association-breaking component by the accessibility of the directed transfer event. If $\omega \to \omega_0 > 0$ near the symmetric state,

$$V_L^{\mathrm{HGT}} \sim \omega_0 \frac{L}{2|S|} \tag{19}$$

and therefore

$$\boxed{\frac{V_L^{\mathrm{HGT}}}{\sqrt{L}} \sim \frac{\omega_0}{\sqrt{8}} \epsilon^{-1/2}.} \tag{20}$$

Maximum configurational entropy is thus the HGT-critical state when transfer remains accessible there.

The accessibility exponent has a parallel interpretation. Define the scaled distance from symmetry $\phi = |S|/\sqrt{L}$ and suppose $\omega(\phi) \sim \omega_0 \phi^\gamma$. Then

$$\frac{V_L^{\mathrm{HGT}}}{\sqrt{L}} \sim \omega_0 2^{(\gamma-3)/2} \epsilon^{-(1-\gamma)/2}. \tag{21}$$

The reciprocal-theory exponent $1/2$ is recovered at $\gamma = 0$; systematic closure of transfer pathways near degeneracy weakens it.

Finite time regularises all of these divergences. At exact degeneracy and constant accessibility,

$$V_L^{\mathrm{HGT}}(T) = \omega_0 \frac{LT}{4}. \tag{22}$$

Figure 2 placeholder.

Panel A: log–log CCDF of simulated or empirical finite-horizon V_{HGT}, with slope -1 for $\gamma = 0$ and steeper curves for $\gamma > 0$.

Panel B: $V_L^{\mathrm{HGT}}/\sqrt{L}$ versus entropy deficit $\mathcal{H}_{\max} - \mathcal{H}$, showing the $-1/2$ critical slope and finite-time rounding.

Panel C: intervention cartoon showing that deleting or down-weighting a few high-score transfer edges truncates the tail more strongly than uniform reduction of all edges with the same mean reduction.

Figure 2: Predicted critical scaling and its epidemiological consequence. Smooth nonzero transfer accessibility preserves the sex-theory exponents. Coupling between accessibility and selective degeneracy changes the exponents in a predictable direction. Because risk is concentrated in the tail, interventions that specifically truncate high-value transfer opportunities can outperform interventions judged only by their effect on mean transfer.

Thus empirical studies should not seek literal divergence. They should seek the predicted scaling range followed by rounding set by observation time, plasmid loss, ecological turnover, spatial separation or other processes that terminate the association.

The two-association model also remains relevant in diverse communities. In a many-genotype system, generic degeneracies at the leading edge are pairwise: equality of two leading fitnesses requires one condition, whereas simultaneous equality of three requires two. Near such a pairwise crossing, faster modes relax and the slow dynamics reduce to the two leading associations [16]. For plasmids, this reduction holds only on edges of the transfer graph that are biologically accessible. HGT therefore adds a graph constraint to the critical theory: pairwise degeneracy is dynamically important only when the corresponding donor–recipient transition has nonzero weight.

4 Direct horizontal transmission and the evolution of conjugation

A plasmid gains from conjugation in two conceptually distinct ways. First, transfer copies the plasmid into another cell. Second, transfer releases the plasmid from its current chromosomal future. Let η denote a plasmid's conjugation investment or efficiency. Over an ecological episode of duration T, write its cumulative log-fitness contribution schematically as

$$\mathcal{G}_T(\eta) = D_T(\eta) + \omega_T(\eta)V_{\mathrm{esc}} - C_T(\eta), \tag{23}$$

where D_T is the direct transmission contribution, C_T is the cumulative cost to vertical success, and $\omega_T V_{\rm esc}$ is the background-escape contribution. For a small modifier change $\delta\eta$,

$$\Delta \log \text{odds}(\eta) \simeq \delta\eta \left[D'_T(\eta) + \omega'_T(\eta) V_{\rm esc} - C'_T(\eta)\right]. \tag{24}$$

Equation (24) makes clear why the heavy tail matters even when conjugation has an ordinary direct transmission benefit. If $V_{\rm esc}$ is heavy-tailed, selection on transfer machinery is episodic: modest costs can dominate most of the time, while rare background-escape episodes generate much larger positive selection coefficients. Evolution need not optimise conjugation for the mean environment.

This perspective is consistent with the empirically observed nonlinear tradeoff between horizontal and vertical plasmid transmission. Across 40 antimicrobial-resistance plasmids from clinical *Escherichia coli*, conjugation efficiency spans a broad range while host growth is relatively tolerant below a threshold; above that threshold, growth costs rise strongly [7]. A convenient phenomenological representation is

$$C(x, a) = C_0 + \frac{J}{2}\left[1 + \tanh\{a(x - x_c)\}\right], \qquad x = \log_{10} \eta, \tag{25}$$

where x_c is a transfer threshold, a controls the sharpness of the cost boundary and J is the high-transfer cost increment. Under a light-tailed distribution of transfer benefits, selection is expected to be governed mainly by typical opportunities. Under a heavy-tailed distribution, a different strategy becomes plausible: remain on a broad low-cost plateau while retaining the capacity to exploit rare high-value transfer windows. This suggests, as a testable hypothesis rather than a theorem, that heavy-tailed escape opportunity can favour regulatory architectures with a sharp transition between low-cost competence and costly over-expression.

5 From evolutionary potential to plasmid epidemiology

5.1 A directed plasmid–host transfer graph

For antibiotic-resistance plasmids, the accessibility factor can be decomposed into measurable processes. Let plasmid j transfer from donor background k to recipient background i. Write the instantaneous successful-transfer rate as

$$\beta_{ijk}(t) = \eta_j(t)\, \chi_{ij}(t)\, \rho_{ik}(t)\, g_{ij}(t), \tag{26}$$

where η_j is conjugation efficiency, χ_{ij} is molecular and physiological permissiveness of recipient i to plasmid j, ρ_{ik} is donor–recipient contact opportunity, and g_{ij} is the probability that a transfer produces an established plasmid–host association. This decomposition is deliberately modular. Host range and restriction systems primarily affect χ and g; community structure affects ρ; transfer-gene regulation affects η; and antibiotics can alter several terms simultaneously by changing donor abundance, recipient abundance, growth and the fitness value of the resistance determinant. Directional transfer bias among bacterial relatives provides one empirical example of structure in these accessibility weights [17].

Clinical data already show that these factors are heterogeneous rather than small perturbations around a mean. The fitness effect of a plasmid varies among bacterial hosts [8], and pOXA-48 conjugation permissiveness and resistance phenotypes vary among clinical enterobacterial backgrounds, with a subset of permissive, high-resistance clones contributing disproportionately to plasmid maintenance [9]. Such heterogeneity is precisely what turns equation (26) into an evolutionary filter on the theoretically available background-escape events.

For each accessible directed edge $e = (k \to i, j)$, define a finite-horizon escape potential

$$V_e(T) = -2q_e\omega_e\Delta X_e\Delta Y_e\frac{\tanh(|U_e|T/2)}{|U_e|}, \tag{27}$$

with the continuous limit at $U_e = 0$. A practical risk score can then combine opportunity and evolutionary value,

$$Q_e(T) = B_e(T)\,[V_e(T)]_+, \qquad B_e(T) = \int_0^T \beta_e(t)N_{D,e}(t)N_{R,e}(t)\,dt, \tag{28}$$

where N_D and N_R denote appropriately scaled donor and recipient abundances and $[z]_+ = \max\{z, 0\}$. B_e estimates how often the transition is attempted and successfully established; V_e estimates how much long-term selective value is exposed by that transition. The exact observation model will differ among gut, hospital and environmental systems, but the distinction between *opportunity* and *value* is general.

5.2 What should be measured?

The theory requires host-resolved plasmid observations, not merely counts of resistance genes. A new resistant clone can arise by clonal spread, by acquisition of the same plasmid into a new bacterial background, or by an independent resistance event. These processes have different meanings in the present framework. Longitudinal recovery of highly similar plasmids from distinct bacterial hosts within a patient provides direct evidence for the background-switching events of interest. For pOXA-48-like plasmids, genomic epidemiology has revealed pervasive within-patient transfer in hospitalised patients [18]; a later UK study reconstructed OXA-48 plasmids from 115 patients and found that identical plasmids commonly occurred in multiple bacterial hosts within the same patient [19]. These observations demonstrate that the required event type is empirically accessible.

Long-read and linked genomic approaches are particularly useful because the theoretical unit is a plasmid–host association. Short-read resistance-gene counts can establish presence but often cannot identify the host or distinguish chromosomal from plasmid contexts. Recent long-read metagenomic methods substantially improve species-resolved host assignment of antibiotic-resistance genes and can identify plasmid-borne contexts [20]. Culture-based isolate sequencing remains valuable when plasmids can be reconstructed and host phenotypes measured. Ideally, surveillance would combine longitudinal host-resolved sequencing with estimates of conjugation, plasmid cost, antibiotic exposure and community abundance.

The fitness decomposition in equation (2) need not be observed perfectly for the framework to be useful. ΔX and ΔY can be treated as latent effects estimated from paired plasmid-carrying and plasmid-cured isolates, competition assays, abundance trajectories, or hierarchical models that borrow information across related hosts. The key empirical quantity is $U = \Delta_X + \Delta_Y$: the theory predicts disproportionate risk when competing accessible plasmid–host associations are close in total fitness but differ substantially in their components. The spin version is even coarser, requiring only the signs and aggregate balance of many component effects.

5.3 From a mean reproductive number to tail risk

Standard plasmid epidemiology asks whether horizontal and vertical transmission are sufficient for persistence, often summarised by a reproduction or invasion threshold [4, 21]. That threshold remains indispensable. The present theory adds a second axis: *how the threshold is approached across heterogeneous evolutionary histories.*

For plasmid j, let $\mathbf{K}^{(j)}$ be a next-generation matrix whose entry $K_{ik}^{(j)}$ is the expected number of established transfers into recipient background i generated by a donor in background k over a specified ecological generation. The plasmid's horizontal invasion number is related to the spectral radius $\rho(\mathbf{K}^{(j)})$. Under heterogeneous hosts and environments, however, $\mathbf{K}^{(j)}$ itself varies among episodes. We therefore define

$$\mathcal{R}_{\mathrm{tail}}(1) = \mathbb{P}(R_{\mathrm{pARG}} > 1)\,, \tag{29}$$

where R_{pARG} is the relevant within-host, between-host or combined plasmid reproduction number under the sampled ecological state. Two plasmids can have the same mean R_{pARG} while having very different $\mathcal{R}_{\mathrm{tail}}(1)$ if one has access to rare high-value transfer states.

The critical theory predicts that the same asymmetry can appear one level below the reproduction number, in the distribution of V_e or Q_e. Consequently, averaging conjugation rates across host backgrounds can discard precisely the observations most informative about future spread. A surveillance programme should estimate not only $\mathbb{E}[\eta]$, $\mathbb{E}[R_{\mathrm{pARG}}]$ or mean plasmid prevalence, but also the upper tail of successful transfer opportunity and the degree to which that tail is concentrated in particular recipient backgrounds or ecological states.

6 Monitoring and thwarting antibiotic-resistance plasmids

The framework suggests a concrete workflow for resistance surveillance.

1. Reconstruct the directed association network. For each sampling interval, identify plasmid or pARG sequence types and their bacterial hosts. Distinguish repeated recovery of a plasmid within one clone from appearance of the same plasmid in a new chromosomal background. The latter defines a candidate HGT edge. Where possible, estimate donor and recipient abundances rather than treating detection as binary.

2. Estimate edge accessibility. For each plasmid–host pair, estimate or infer the factors in equation (26): conjugation efficiency, recipient permissiveness, donor–recipient opportunity and establishment. Laboratory mating assays can calibrate $\eta\chi g$ for representative pairs, whereas time-resolved abundance and co-occurrence data inform ρ. Experimental estimates should follow explicit reporting and modelling standards because conjugation-rate estimates are sensitive to assay assumptions [22].

3. Estimate finite-time escape value. Use plasmid-cured controls, paired isolates, growth assays or latent-effect models to estimate the plasmid and host components of fitness. Compute $V_e(T)$ rather than the infinite-time expression, with T set by the epidemiologically relevant window: gut colonisation, ward stay, antibiotic course, environmental residence or another clearly defined episode. This automatically regularises the critical singularity.

4. Fit the tail, not only the centre. Compare the empirical distribution of positive V_e or Q_e with the $1/v$ null prediction and alternatives implied by equation (13). A slope near -1 supports smooth accessibility through the cancellation surface. A steeper tail can indicate that high-value states are progressively inaccessible; a hard cutoff can indicate strong host-range barriers, rapid plasmid loss or short ecological duration. Light-tailed families such as exponential or gamma distributions provide useful null alternatives, while truncated Pareto, lognormal and Weibull families can describe finite biological tails.

5. Rank interventions by tail truncation. Equation (26) identifies four direct levers: reduce transfer efficiency η, reduce recipient permissiveness χ, reduce effective contact ρ, or reduce establishment g. Plasmid loss and shortened duration provide additional ways to truncate K_T. Antibiotic use changes the fitness components themselves and can move the system towards or away from cancellation while also amplifying resistant donors and recipients. The theory therefore does not prescribe a universal intervention. It provides a ranking principle: prefer changes that most reduce the upper tail of Q_e or $\mathbb{P}(R_{\mathrm{pARG}} > 1)$ per unit cost, rather than changes that merely reduce mean conjugation.

This last distinction is potentially consequential. If Q_e is heavy-tailed, a small number of plasmid–host transitions can dominate cumulative risk. Uniformly reducing all transfer rates by a modest factor may leave those transitions above threshold. Conversely, an intervention that removes a small set of high-permissiveness recipients, interrupts their contact opportunities, or sharply reduces establishment in those backgrounds can truncate the tail even if the community-wide mean transfer rate changes little. In graph terms, the target is not necessarily the highest-degree node; it is the edge or node with the greatest contribution to the tail of evolutionary amplification.

The proposal is compatible with several existing control approaches without depending on any one of them. Infection-control measures can reduce ρ across patients or wards. Antibiotic stewardship can alter the selective value and abundance of resistance plasmids, although its effects on microbial community structure must be modelled rather than assumed. Experimental anti-conjugation strategies act on η; plasmid-curing or incompatibility-based approaches act on persistence or establishment. The theory supplies a common currency in which their effects can be compared: how strongly each intervention truncates high-value background escape and reduces the probability of supercritical plasmid spread.

7 Testable predictions

The synthesis yields several predictions that distinguish it from a generic statement that HGT is beneficial.

First, across sufficiently heterogeneous evolutionary episodes with smooth transfer accessibility, the positive tail of background-escape potential should approach

$$\mathbb{P}(V_{\mathrm{esc}} > v) \propto v^{-1}. \tag{30}$$

The prediction concerns the CCDF, not the probability density. Finite time and finite population size should truncate the tail at large v without changing its intermediate asymptotic slope.

Second, if host-range or establishment barriers close systematically near selective degeneracy, the tail should steepen according to equation (13). This provides an empirical route to estimating the accessibility exponent γ from transfer data. The same γ should weaken the entropy singularity in the binary representation according to equation (21).

Third, successful transfers should be enriched among donor–recipient pairs whose *total* fitnesses are similar but whose component plasmid and host effects are antagonistic. Equality of total fitness alone is not sufficient: if all components are small, little latent value exists to release. The predicted high-risk state is cancellation of substantial opposing effects.

Table 1: Quantities linking the HGT theory to resistance surveillance.

Quantity	Interpretation	Possible empirical source
η_j	Conjugation efficiency of plasmid j	Standardised mating assays; transfer-gene expression; calibrated literature values
χ_{ij}	Recipient permissiveness	Pairwise donor–recipient assays; host-range screens; restriction/incompatibility genotype
ρ_{ik}	Donor–recipient opportunity	Co-abundance, spatial co-localisation, patient/ward contact, microbiome time series
g_{ij}	Post-transfer establishment	Transconjugant persistence; plasmid loss; growth and competition experiments
$\Delta X, \Delta Y$	Plasmid and host fitness components	Plasmid-cured versus carrying isolates; reciprocal background assays; hierarchical latent-effect models
$V_e(T)$	Finite-time background-escape potential	Computed from fitness components and observation horizon
$Q_e(T)$	Opportunity-weighted escape risk	Combination of successful-transfer exposure and $V_e(T)$
$\mathcal{R}_{\text{tail}}(1)$	Probability of supercritical pARG spread	Episode-specific next-generation models fitted to longitudinal genomic and epidemiological data

Fourth, upper-tail metrics should predict plasmid emergence better than mean conjugation alone when heterogeneity is strong. In prospective data, models containing tail mass of V_e or Q_e should outperform otherwise matched models using only mean η or mean transfer count in predicting newly observed plasmid–host associations.

Fifth, targeted removal or suppression of the highest-Q_e transitions should reduce $\mathcal{R}_{\text{tail}}(1)$ more than a uniform intervention producing the same reduction in mean edge weight. This is a direct experimental or simulation test of the proposed control principle.

Finally, if heavy-tailed escape opportunity shapes the evolution of transfer regulation, plasmids exposed to broader and more heterogeneous host communities should show stronger evidence of operating near a conjugation–growth threshold and, potentially, sharper regulatory transitions than plasmids evolving in narrowly restricted host environments. The nonlinear conjugation–growth tradeoff documented empirically provides an immediate system in which to test this prediction [7].

8 Discussion

The evolutionary literature often treats sex and HGT as related forms of genetic exchange, but the analogy can become either too loose to be predictive or too literal to accommodate bacterial biology. The framework developed here aims for a middle ground. It isolates one common operation—escape from a genetic background—and then asks which mathematical consequences survive the transition from reciprocal recombination to directional plasmid transfer.

Three survive with little modification. First, opposing additive fitness components create negative association and hence latent value that can be released by genetic exchange. Second, when total fitness differences nearly cancel, the competing associations coexist for anomalously long times, causing the integrated value of exchange to become large. Third, continuous sampling across this cancellation surface produces a heavy positive tail. These are scale-level statements: they do not require a particular microscopic source of fitness variation, epistasis, fluctuating selection or mutational process.

HGT adds three biologically important complications. The first is directionality. Conjugation does not generally create a reciprocal pair of recombinants, so its amplitude depends on which transfer direction is available. The second is direct reproduction. A plasmid can benefit simply because conjugation makes another plasmid-bearing cell; background escape is an additional effect, not a substitute for ordinary horizontal transmission. The third is accessibility. Host range, restriction, incompatibility, contact and establishment define a directed graph on which only some theoretically possible recombinants can be realised. The accessibility exponent γ summarises how this graph intersects the selective cancellation surface.

That final point may be the most useful theoretical addition. Universality is often misunderstood as a claim that details do not matter. Here the opposite is true. The baseline theory identifies which details are *relevant* to the large-scale exponent. Smooth changes in transfer probability alter only the tail amplitude. A systematic zero of accessibility at the cancellation surface changes the exponent or removes the singularity. This is a renormalisation-style distinction that can be tested with plasmid host-range data.

The theory also changes how one might think about antibiotic-resistance surveillance. Genomic epidemiology increasingly resolves plasmids across bacterial hosts within patients and hospitals [18, 19]. The immediate temptation is to count transfers and estimate an average rate. Our analysis argues that rate alone is insufficient when the selective value of a successful background switch is heavy-tailed. A rare transfer into a particularly permissive and selectively favourable host can matter far more than many transfers among backgrounds with little long-term amplification. The natural surveillance object is therefore a joint distribution over transfer opportunity and post-transfer evolutionary value.

Several limitations are important. We have treated plasmid and host contributions as additively separable for the purpose of defining the large-scale cancellation coordinate. Real plasmid–host interactions are epistatic, and epistasis can itself vary systematically among hosts [23]. The scale argument predicts that much of this contingency can be absorbed into effective component effects, but strong structured epistasis could correlate with accessibility and therefore alter γ. Likewise, the two-lineage reduction assumes that generic slow events are pairwise. Dense polymorphism, frequency-dependent selection, ecological feedback and simultaneous transfer among many comparable hosts can create genuinely higher-dimensional critical dynamics. These cases should be treated as extensions rather than forced into the two-state model.

The surveillance proposal also requires better empirical definitions of an HGT "episode". In the theory, finite time sets the upper cutoff of the singularity. In a patient, the appropriate endpoint might be clearance, discharge, antibiotic cessation or loss of the donor–recipient ecological overlap. In a hospital, it might be ward-specific circulation over a defined interval. These choices change amplitudes and cutoffs but should not change the intermediate scaling if the underlying mechanism is correct.

Finally, the model is a theory of evolutionary risk, not a claim that all plasmid transfer should be suppressed. Plasmids participate in microbial adaptation far beyond antibiotic resistance, and interventions that alter microbial communities can have unintended consequences. The practical value of the framework is prioritisation: identify the plasmid–host transitions whose combination of accessibility, persistence and background-escape value places them in the extreme tail, and focus experimental and epidemiological attention there.

9 Conclusions

Plasmid-mediated HGT can be viewed as a directional, copying form of genetic background escape. Once the direct reproductive benefit of conjugation is separated from this escape component, recent scale-level theories of sex extend with surprisingly few changes. Near cancellation of opposing plasmid and host fitness effects, the background-escape value of transfer becomes large because favourable covariance structure and long persistence coincide. Smooth transfer accessibility preserves a $1/v$ tail in episode-integrated HGT advantage and a $-1/2$ entropy singularity in the binary spin representation. HGT-specific host-range structure modifies these exponents only when accessibility itself vanishes near the critical surface.

This distinction turns a general evolutionary argument into a surveillance hypothesis. Host-resolved plasmid sequencing can reveal the directed transitions; laboratory and ecological data can estimate their accessibility; fitness decomposition can estimate finite-time escape value; and epidemiological models can ask whether the upper tail, rather than the mean, predicts supercritical pARG spread. If so, resistance control becomes a problem of tail truncation: prevent the rare background switches that expose unusually large evolutionary value before they seed the next episode of spread.

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